

Isometry

Angle between vectors:

$$\cos(\theta) = \frac{u \cdot v}{|u| \cdot |v|}$$

Lines:

$$y = mx + c \text{ or } PQ = \lambda v$$

Transformation: A unique (u, v) such that $(u, v) = \alpha(x, y)$

Inverses: Inverses exist if α is a transformation, then,

$$\alpha^{-1} = (x(u, v), y(u, v))$$

Translation:

$$\tau_v(x, y) = (x + v_1, y + v_2)$$

Half-Turn:

$$\eta_Q(x, y) = (2q_1 - x, 2q_2 - y)$$

Isometry:

$$d(P, Q) = d(\alpha(P), \alpha(Q))$$

Reflection:

$$\sigma_l(P) = P' \text{ where } PP' = -2(QP - u^\perp) \cdot u^\perp$$

Rotation:

$$\rho_{P, \theta}(P) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} P$$

Fixed Points: P is a fixed point if for a transformation, $\alpha(P) = P$

Fixed points for known transformations:

- $\eta_Q \rightarrow Q$
- $\tau_v \rightarrow \emptyset$
- $\sigma_l \rightarrow l$
- $\rho_{Q, \theta} \rightarrow Q$

Translation as reflection: For PQ connecting lines l and m, and $PQ \perp m$

$$\sigma_l \sigma_m = \tau_{2PQ}$$

Glide Reflection: A reflection across l followed by a translation of v (or a translation followed by a reflection)

$$\gamma_{l, v} = \tau_v(\sigma_l) = \sigma_l(\tau_v)$$

Isometries can be represented as reflections: Every isometry can be presented by at most 3 reflections

- $i = \sigma_l(\sigma_l)$
- $\sigma_l = \sigma_l$
- $\rho_{P, \theta} = \sigma_l(\sigma_m)$, angle between l and m = $\frac{\theta}{2}$

- $\tau_v = \sigma_l(\sigma_m)$, l and m parallel and vector connect the two is $\frac{v}{2}$
- $\gamma_{l,v} = \tau_v(\sigma_l) = \sigma_n(\sigma_m(\sigma_l))$

Even and Odd isometries: Even/Odd if it can be represented by an even/odd number of reflections

Symmetries: α is a symmetry for a set S if $\alpha(S) = S$ The symmetry group of a shape is the set of all ρ and σ that allow the shape to look the same

Affine and Projective Geometry

Affine Transformation: An affine transformation has the form $Ax + b$. The derivative of an affine transformation is just A .

Orthogonal: A matrix is said to be orthogonal is $A^{-1} = A^T$. An affine transformation is an isometry if and only if its derivative is orthogonal

Projective Plane: A point on the projective plane is represented as $(a : b : c)$ which is equivalent to $(a\lambda, b\lambda, c\lambda)$ Euclidean plane is a subset of the projective plane, $\forall (a, b) \in \mathbb{R}^2, (a : b : 1) \in \mathcal{P}$ Lines in

Projective Plane: A line in the projective plane is $\langle a : b : c \rangle \Leftrightarrow a\lambda x + b\lambda y + c\lambda z = 0 \mid l \cap E_1 \Leftrightarrow ax + by + c = 0 \Leftrightarrow (x : y : 1)$ Line at infinity: $l_\infty = \langle 0 : 0 : 1 \rangle$ equivalent to $z = 0$ Line in

Projective Plane: Line passing through $P = (p_1, p_2, p_3), Q = (q_1, q_2, q_3)$ is $\det \begin{pmatrix} x & y & z \\ p_1 & p_2 & p_3 \\ q_1 & q_2 & q_3 \end{pmatrix}$

Collineation: A collineation is a transformation that maps lines to lines. It is represented by a matrix A . With 4 points, P, Q, R, S and P', Q', R', S' then we there exists a unique collineation that maps each point.

Polynomials: degree = sum of highest power, homogenous if all terms are of the same degree

Conics

Conics: A homogenous polynomial of degree 2, F , is called a conic if $F(x, y, z) = 0$. Conics have a matrix form of $x^T M x = 0$. Tangent to a Conic: The tangent line to a conic with matrix M , at point x_0 is the line $x_0^T M x = 0$. This can either be a line or just the point p (if the conic is degenerate).

Transformations of Conics: If C is a conic and φ is a collineation, then $\varphi(C)$ is another conic.

Intersection with Euclidean plane: $C \cap l_\infty = \emptyset$, point, 2 points if the result is an ellipse, parabola, hyperbola.

Graphs

Types of graphs:

- C_n cyclic graphs
- P_n path graphs
- K_n complete graphs
- $K_{n,m}$ complete bipartite graphs

Degree: The degree of a vertex is the number of connected edges

Vertex Degree Equation: sum of all degrees of every vertex = 2 times the number of edges

Euler Characteristic of a Graph: For a graph $G(V, E)$, then the Euler Characteristic is $\chi(G) = |V| - |E|$

Circuits or Cycles: A circuit is a path that ends at the same index as it starts. A circuit is irreducible if it does not contain the same vertex more than once.

Trees: A tree contains no cycles. A leaf on a tree has degree 1. $\chi(T) = 1$ for any tree T .

Spanning Tree: A spanning tree of a graph G , contains all vertices of G but forms a tree.

Eulerian Circuit: A Eulerian circuit passes every edge exactly once. A graph is Eulerian if and only if every vertex has even degree. A graph can have a Eulerian path if it has exactly 2 vertices of odd degree. (start, end vertices).

Surfaces

Euler Characteristic for Surfaces: $\chi(S) = |V| - |E| + |F|$

Polygonal Decomposition: A surface can be turned into a word, a sequence of vectors along the surface. Homeomorphic surfaces have the same polygonal decomposition.

Connecting Surfaces: 2 surfaces, S and T can be connected with a cylinder. Then we can describe the connected surface as $S \# T$ with the property that $\chi(S \# T) = \chi(S) + \chi(T) - 2$ (-2 comes from cutting out 2 disks).

Classification of Surfaces: A surface S can be defined by $S \cong S^2 \#^d D^2 \#^p P^2 \#^t T$, which is the standard form of a surface. That means surfaces are unique for a unique set of (d, p, t) up to homeomorphism. Using a surface in standard form, we can say $\chi(S) = 2 - d - p - 2t$ Graphically, d = number of punctures, t = number of handles, p = number of caps.

Algorithm for Standard form of Surface:

1. orientable if edges are paired $(a, \text{reversed } a)$, if orientable, $p=0$, otherwise $t=0$
2. find number of loops formed by unpaired edges, d = number of loops
3. use Euler characteristic to find p or t

Dual surface: Swap faces and vertices.

Planar: A graph is planar if it has no subgraph isomorphic to a subdivision of K_5 or $K_{3,3}$. A connected planar graph has a $\chi(S) = 2$